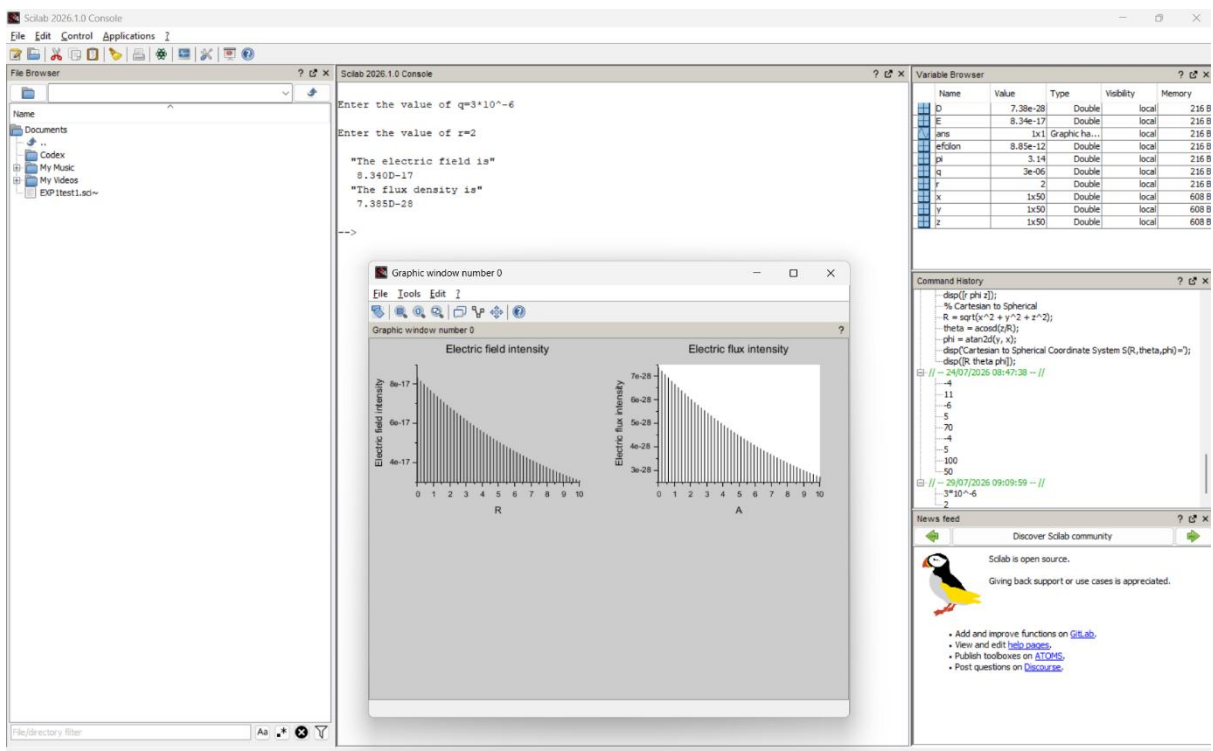


EXP 2: Determination and plotting of Electric field and Electric flux density in sphere full of charges with varying radius.

```

1 clear;
2 clear;
3 Q=input('Enter the value of q=');
4 r=input('Enter the value of r=');
5 epsilon=8.854*10^-12;
6 pi=3.14;
7 E=q/(pi*epsilon*r^2);
8 disp('The electric field is', [E]);
9 D=E*epsilon;
10 disp('The flux density is', [D]);
11 x = linspace(0, 10, 50);
12 y = E * exp(-0.1 * x);
13 z = D * exp(-0.1 * x);
14 figure();
15 subplot(2, 2, 1);
16 plot(x,y);
17 xlabel('R');
18 ylabel('Electric field intensity');
19 title('Electric field intensity');
20 subplot(2, 2, 2);
21 plot(x,z);
22 xlabel('A');
23 ylabel('Electric flux intensity');
24 title('Electric flux intensity');

```



Exp 2:-

$$Q = 3 \times 10^{-6}$$

$$r = 2 \text{ mm}$$

output

The electric field is $= 8.340 \times 10^{-17}$

The flux density is $= 7.385 \times 10^{-21}$

Electric field

$$E = \frac{Q}{4\pi\epsilon_0 r^2}$$

$$E = \frac{3 \times 10^{-6}}{4} \times 3.14 \times 8.854 \times 10^{-12} \times (2 \times 10^{-3})^2$$

$$= 2.357 \times 10^{-6} \times 8.854 \times 10^{-12} \times 4 \times 10^{-6}$$

$$= 2.085 \times 10^{-10} \times 4 \times 10^{-7}$$

$$E = 8.340 \times 10^{-17}$$

electric flux density

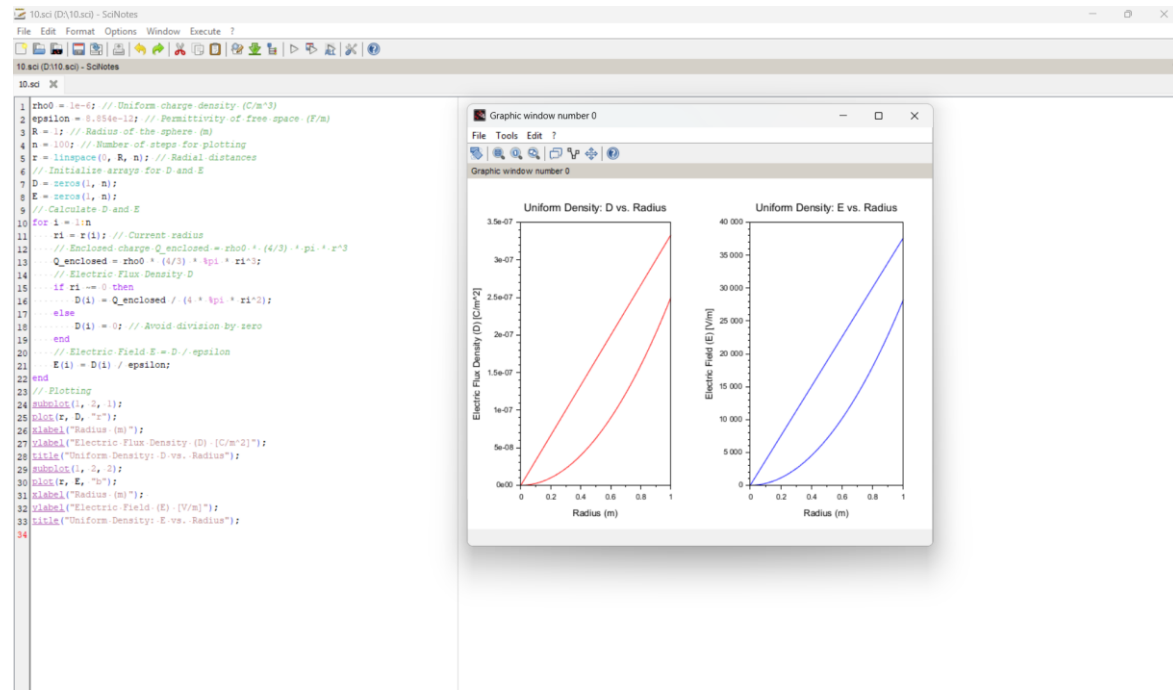
$$D = \epsilon E$$

$$= (8.854 \times 10^{-12}) (8.340 \times 10^{-17})$$

$$D = 7.385 \times 10^{-21}$$

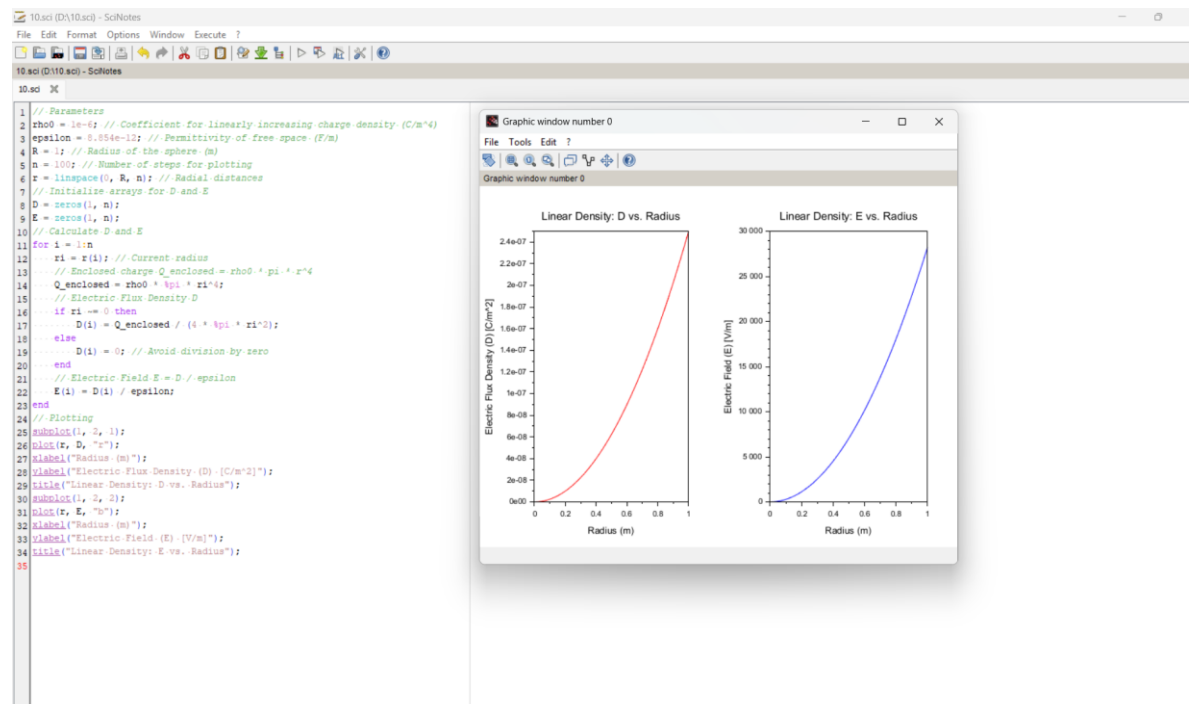
Case 1:

Derive and plot the electric field (E) and electric flux density (D) inside a uniformly charged sphere with charge density ρ_0 . What are the values of E and D at the center and at the surface of the sphere?



Case 2:

For a sphere with a linearly increasing charge density $\rho_v = \rho_0 r$, derive the expressions for the electric field (E) and electric flux density (D). Plot their variations inside the sphere and analyze the trends.



Case 3:

A sphere of radius $R=2$ m has a uniform charge density $\rho=3\times 10^{-6}$ C/m³

1. Calculate and plot the electric flux density $D(r)$ and electric field $E(r)$ as a function of radial distance r , ranging from 0 to 4 meters.
2. Discuss the behaviour of the electric field inside and outside the sphere.

